R Programming Practical Examination

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2026-11-11

Part A: Basic Operations

A1

```
v <- seq(5:50)
length(v)
```

```
## [1] 46
```

```
sum(v)
```

```
## [1] 1081
```

```
mean(v)
```

```
## [1] 23.5
```

A2

```
w <- rep(c(1, 2, 3))
rep(c(1, 2, 3), times = 4)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z <- rep(c(1, 2, 3))
rep(c(1, 2, 3), each = 4)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```
remainder <- 17 %% 15
integer_division <- 17 %/% 15
power <- 2^10
square_root <- sqrt(144)
pi
```

```
## [1] 3.141593
```

```
remainder
```

```
## [1] 2
```

```
integer_division
```

```
## [1] 1
```

```
power
```

```
## [1] 1024
```

```
square_root
```

```
## [1] 12
```

```
pi
```

```
## [1] 3.141593
```

A4

```
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)

a + b
```

```
## [1]  3  7 11 15
```

```
a * b
```

```
## [1]  2 12 30 56
```

```
a > b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
sum(a * b)
```

```
## [1] 100
```

A5

```
paste("Student-1", "Student-2", "Student-3")
```

```
## [1] "Student-1 Student-2 Student-3"
```

```
typeof(42)
```

```
## [1] "double"
```

```
typeof(42L)
```

```
## [1] "integer"
```

```
typeof("42")
```

```
## [1] "character"
```

```
typeof(41i)
```

```
## [1] "complex"
```

```
typeof(TRUE)
```

```
## [1] "logical"
```

##Part B - Data frames and Conditional Statements B1

```
students <- data.frame(
id = 1:8,
name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
section = c("A", "B", "A", "C", "B", "C", "A", "B"), midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
final = c(82, 79, 96, 71, 80, 91, 93, 65),
attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
```

```
stringsAsFactors = FALSE)
students
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

B2

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
students$average <- round((students$midterm + students$final) / 2, 1)
students$average
```

```
## [1] 80.0 82.0 94.0 69.0 77.0 89.5 94.0 62.0
```

B4

```
students$passed <- (students$average >= 70) & (students$attendance >= 75)
students$passed
```

```
## [1]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE
```

B5

```
students$standing <- ifelse(students$average >= 90, "Honors",
 ifelse(students$average >= 75, "Good", "Needs support"))

students$standing
```

```
## [1] "Good"          "Good"          "Honors"        "Needs support"
## [5] "Good"          "Good"          "Honors"        "Needs support"
```

B6

```
classify <- function(score){
  if (score >= 85){
    return("Distinction")
  } else if (score >= 70){
    return("Merit")
  } else if (score >= 50){
    return("Fail")
  }
}
print(sapply(c(90, 73, 55, 40), classify))
```

```
## [[1]]
## [1] "Distinction"
##
## [[2]]
## [1] "Merit"
##
## [[3]]
## [1] "Fail"
##
## [[4]]
## NULL
```

##Part C - rbind() and cbind() C1

```
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE )
new_students$average <- round((new_students$midterm + new_students$final) / 2, 1)
new_students$passed <- (new_students$average >= 70) & (new_students$attendance >= 75)
new_students$standing <- ifelse(new_students$average >= 90, "Honors", ifelse(new_students$average >= 75

new_students
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 2 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

```
new_students$average
```

```
## [1] 82.5 69.0
```

```
new_students$passed
```

```
## [1]  TRUE FALSE
```

```
new_students$standing
```

```
## [1] "Good"          "Needs Support"
```

```
students_all <- rbind(students, new_students)
```

C2

```
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
print(class(students_all))
```

```
## [1] "data.frame"
```